Pair Production, Wave–Particle Duality, and Matter Waves

PHYS 2203 — Introductory Physics III

October 23, 2026

Overview

The photoelectric effect (Lecture 22) and the Compton effect (Lecture 23) established that a photon carries energy $E = hf$ and momentum $p = h/\lambda$. This lecture first shows that a high-energy photon can turn into an electron and a positron, using the mass–energy relations and invariant mass of Lectures 18 and 19. The double-slit experiment performed one photon at a time then shows that light behaves both as a wave and as a particle. Repeating it with electrons shows that matter does the same. We then see how crystals diffract electrons according to the Bragg law, and how the wavelength of a matter wave is set by the de Broglie relation $\lambda = h/p$.

1 Pair Production

In 1932 Carl Anderson, studying cosmic rays, found a particle with the same mass as the electron but positive charge: the **positron**, the antiparticle of the electron. (He received the 1936 Nobel Prize.) A high-energy photon passing near a nucleus can convert into an electron–positron pair,

$$\gamma \longrightarrow e^- + e^+, \tag{1}$$

which is called **pair production**. Charge is conserved, since the pair has total charge zero. In a magnetic field the electron and positron curve in opposite directions, which is how pairs are identified in particle detectors (Figure 1).

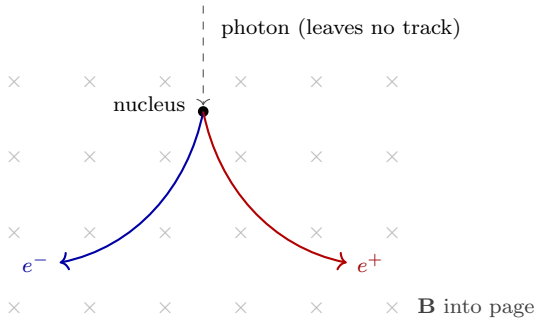


Figure 1: Pair production near a nucleus. The photon is uncharged and leaves no track, and the electron and positron curve in opposite directions in the magnetic field.

The threshold energy

The photon must supply at least the rest energy of the two particles it creates. The least energetic photon that can produce a pair leaves both particles with no kinetic energy:

$$\boxed{\frac{hc}{\lambda} \geq 2m_e c^2 = 2(0.511 \text{ MeV}) = 1.022 \text{ MeV}.} \tag{2}$$

The corresponding wavelength is

$$\lambda = \frac{hc}{2m_e c^2} = \frac{1240 \text{ eV} \cdot \text{nm}}{1.022 \times 10^6 \text{ eV}} = 1.21 \times 10^{-3} \text{ nm} = 1.21 \text{ pm}, \tag{3}$$

shorter than X-ray wavelengths. Electromagnetic radiation in this range is called **gamma rays** (γ).

Why a nucleus is needed

A photon cannot turn into a pair in empty space, because energy and momentum cannot both be conserved. The cleanest way to see this uses the invariant mass of Lecture 19,

$$M_{\text{sys}}^2 c^4 = \Big(\sum E\Big)^2 - \Big(\sum p\Big)^2 c^2, \tag{4}$$

which is conserved and has the same value in every frame. For a single photon, $E = pc$, so $M_{\text{sys}} = 0$. For an electron–positron pair, evaluate it in the frame where the pair's total momentum is zero: the total energy there is at least $2m_e c^2$, so $M_{\text{sys}} \geq 2m_e$. Since $0 \neq 2m_e$, the process is impossible on its own.

A nearby nucleus solves the problem by absorbing momentum. Because the nucleus is heavy, it takes very little energy in doing so. Suppose a lead nucleus ($Mc^2 = 1.94 \times 10^5$ MeV) takes up all the momentum of a photon at threshold, $p = E/c = 1.022$ MeV/c. Its kinetic energy is

$$K = \frac{p^2}{2M} = \frac{(pc)^2}{2Mc^2} = \frac{(1.022 \text{ MeV})^2}{2(1.94 \times 10^5 \text{ MeV})} = 2.7 \times 10^{-6} \text{ MeV} = 2.7 \text{ eV}, \tag{5}$$

negligible compared with 1.022 MeV. The nucleus carries off the momentum, and essentially all of the photon's energy goes into the pair. A lighter partner would not work as well. It would carry off a larger share of the energy as kinetic energy, so the photon would need more than 1.022 MeV. With a free electron as the partner, the threshold rises to $4m_e c^2 = 2.04$ MeV.

2 Wave or Particle?

We now have two descriptions of light. Interference and diffraction, studied earlier in the course, show that light is a wave. The photoelectric effect, the Compton effect, and pair production show that light is made of particles with energy hf and momentum h/λ. Which is correct?

Which behavior we see depends on the experiment, and in particular on how the wavelength λ compares with the relevant size D of the apparatus. When λ is comparable to or larger than D, as for light passing through a narrow slit, wave behavior such as diffraction appears. When λ is much smaller than D, as for an X-ray photon striking a single electron, the radiation acts as a localized particle.

The double-slit experiment, one photon at a time

The double-slit experiment shows both behaviors at once. With bright light, the screen shows the familiar interference pattern of bright and dark fringes. Now reduce the intensity until photons pass through the apparatus one at a time, and replace the screen with a detector that records where each photon arrives.

Each photon is detected at a single point, like a particle. The first few arrivals look randomly scattered. But as more photons accumulate, the interference pattern appears (Figure 2). Photons arrive often where the wave theory predicts bright fringes and never where it predicts dark fringes, even though only one photon is in the apparatus at a time.

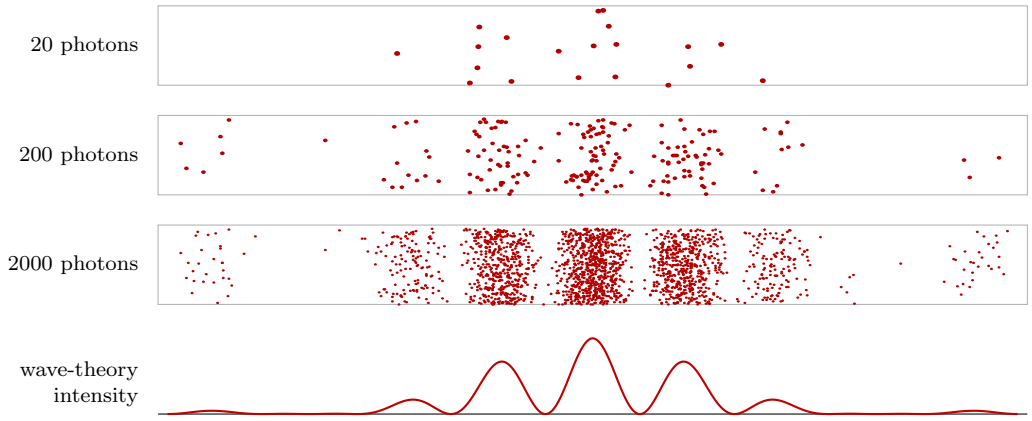


Figure 2: Simulated double-slit pattern built up one photon at a time. Each photon arrives at a single point, but the accumulated arrivals reproduce the interference pattern predicted by wave theory (bottom).

We cannot predict where any particular photon will land. What we can predict is the probability of finding it in a given region, and this follows the intensity of the classical wave, which is proportional to the square of the wave's amplitude:

Probability and the wave. The probability of detecting a photon in a small region is proportional to the intensity of the electromagnetic wave there, that is, to the square of the amplitude of the wave in that region.

With bright light, enormous numbers of photons arrive, as in the laser-pointer example of Lecture 22, and the random arrivals average out to the smooth classical intensity. This is an example of the **correspondence principle**: a new theory must agree with the old one in the regime where the old one is known to work. Here the wave description of light is recovered when very many photons are involved.

Light is therefore neither simply a wave nor simply a particle. It travels and interferes like a wave, and it is detected in discrete particles. This is called **wave–particle duality**. As we see next, it is not special to light.

3 Matter Waves: The Double Slit with Electrons

If light, classically a wave, can behave as particles, can matter, classically made of particles, behave as waves? Repeat the double-slit experiment with a beam of electrons of a single energy, sent through slits so slowly that only one electron is in the apparatus at a time. A screen records a small flash where each electron arrives (Figure 3).

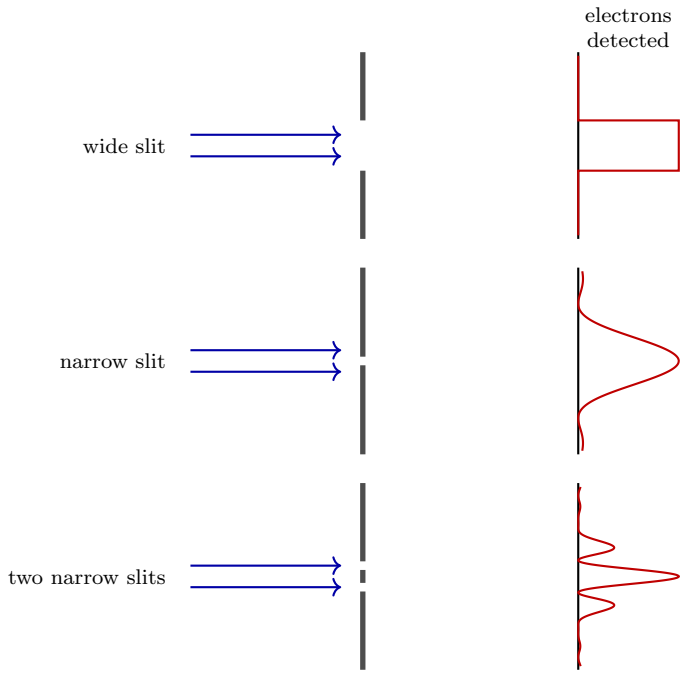


Figure 3: Electrons sent one at a time through a wide slit, a narrow slit, and two narrow slits. The red curves show where electrons accumulate on the screen. With two slits open, there are points where no electron ever arrives, although electrons did arrive there with either slit open alone.

- **Wide slit:** the electrons pass straight through and hit a stripe the width of the slit, as particles should.
- **Narrow slit:** the electrons arrive one by one all over the screen. The beam spreads out, like a wave diffracting.
- **Two narrow slits:** the arrivals build up an interference pattern, exactly like Figure 2 for photons. Opening the second slit makes certain points on the screen receive *no* electrons at all, even though they received electrons with either slit open alone.

A particle going through one slit would have no reason to avoid a point just because another slit was opened. The zeros are destructive interference, which requires two waves. So each electron behaves as a wave that passes through *both* slits and interferes with itself, and is then detected at a single point, like a particle. Electrons show the same wave–particle duality as photons.

The associated wave is called a **matter wave**, described by a **wave function** $\Psi(x,t)$. As for light, the wave gives the probability of where the particle is detected:

Probability and the matter wave. The probability of finding a particle in a small region is proportional to $|\Psi|^2$, the square of the amplitude of its matter wave in that region.

For this reason Ψ is also called the **probability amplitude**. What oscillates in a matter wave is not the particle or its mass, but this probability amplitude.

4 Electron Diffraction and the Bragg Law

Matter waves have very short wavelengths, so double slits for electrons are extremely hard to make. Nature provides a better device: a crystal, whose atoms are regularly spaced about 0.1–0.3 nm apart. Each atom scatters an incoming wave in all directions, so the crystal acts like a three-dimensional diffraction grating. In 1927 Davisson and Germer found that electrons scattered from a nickel crystal come off strongly only at certain angles, the first direct evidence that matter has a wave nature.

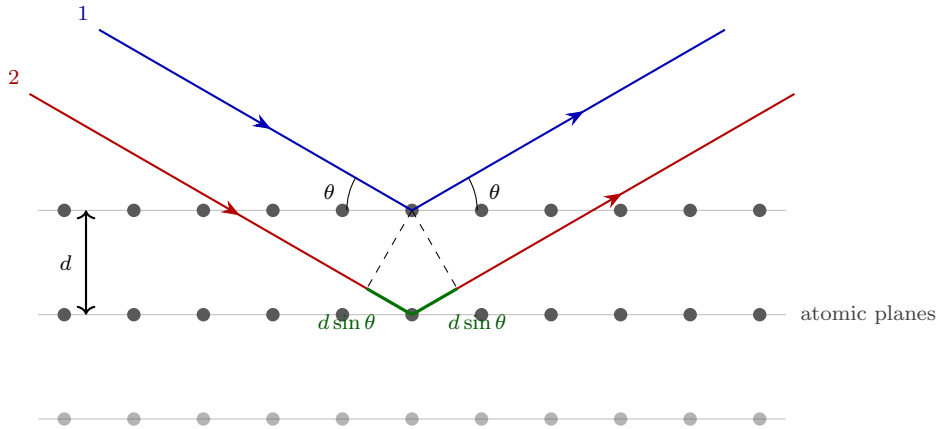


Figure 4: Waves scattered from two neighboring atomic planes a distance d apart. The angle θ is measured from the planes. Ray 2 travels an extra distance $2d\sin\theta$ (the two green segments) compared with ray 1.

A wave strikes the crystal at angle θ to a set of atomic planes, and the detector receives the wave leaving at the same angle. Waves scattered by atoms in the same plane all travel the same distance, so the interference is between different planes. The wave scattered by the next plane down travels an extra distance $2d\sin\theta$. Constructive interference requires this to be a whole number of wavelengths:

$$\boxed{2d\sin\theta = m\lambda, \qquad m = 1, 2, 3, \ldots} \tag{6}$$

This is the **Bragg law**, found by W. H. and W. L. Bragg (1915 Nobel Prize) for X-rays. The same law applies to electrons, neutrons, and even whole atoms, because a wave is a wave. Note that θ is measured from the *planes*, not from the normal.

Example: the Davisson–Germer experiment. In their nickel crystal the relevant atomic planes are $d = 0.091$ nm apart. Electrons with kinetic energy 54 eV gave a strong peak at $\theta = 65°$. Taking $m = 1$,

$$\lambda = 2d\sin\theta = 2(0.091\text{ nm})\sin 65° = 0.165\text{ nm}. \tag{7}$$

The electrons diffract as waves with this wavelength. What determines it?

5 The de Broglie Wavelength

In 1924 Louis de Broglie proposed that a particle of momentum p has a matter wave of wavelength

$$\boxed{\lambda = \frac{h}{p}} \tag{8}$$

the same relation as for a photon (Lecture 23). This is the **de Broglie wavelength**. It holds for all particles, at any speed, with the relativistic momentum when needed. De Broglie received the 1929 Nobel Prize.

Example: why we don't notice matter waves. Compare an electron moving at 900 m/s with a 0.145 kg baseball thrown at 40 m/s (about 90 mph):

$$\lambda_{\text{electron}} = \frac{6.63 \times 10^{-34}\ \text{J}\cdot\text{s}}{(9.11 \times 10^{-31}\ \text{kg})(900\ \text{m/s})} = 8.1 \times 10^{-7}\ \text{m}, \tag{9}$$

$$\lambda_{\text{baseball}} = \frac{6.63 \times 10^{-34}\ \text{J}\cdot\text{s}}{(0.145\ \text{kg})(40\ \text{m/s})} = 1.1 \times 10^{-34}\ \text{m}. \tag{10}$$

The electron's wavelength is comparable to that of light, and its wave nature can be revealed. The baseball's is about 19 orders of magnitude smaller than a nucleus, so it always behaves as a particle. Planck's constant is so small that only very light particles have noticeable wavelengths.